PORTFOLIO

Mohamed El Mehdi **MAKHLOUF**

AI/ML & DATA R&D ENGINEER · PHD CANDIDATE · AVAILABLE MAR. 2027

Lyon area, France · Open to opportunities in France and internationally

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Profile

AI/ML and Data R&D engineer completing a PhD in computer science, with expertise in self-supervised and robust learning, signal and time-series analysis, and statistical modelling. End-to-end experience from problem framing and literature review to datasets, simulators, documented APIs and reproducible Python/PyTorch pipelines on national HPC infrastructure. Applied R&D across telecommunications, energy and healthcare; two international conference publications and experience in European collaborative projects. Available from March 2027.

Professional experience

Inria Lyon — CHASER and INSTINCT collaborative projects

Lyon area, France

PHD CANDIDATE AND MACHINE LEARNING RESEARCH ENGINEER — *Mar. 2024 — Feb. 2027*

- Designed self-supervised methods that turn high-dimensional radio measurements into spatial and dynamic representations for pseudo-localisation and future channel-state anticipation.
- Built reproducible Python/PyTorch pipelines covering data preparation, training, metrics, ablation studies, visualisation and execution on Grid'5000 and Jean Zay HPC systems.
- Investigated channel-chart robustness under reconfigurable intelligent surfaces (RIS) and developed invariant methods for downstream radio mapping and control tasks.
- Delivered a multi-cell dataset with generator/simulator and source code, channel-charting applications and API, plus privacy risk analysis and protection mechanisms.
- Reported progress regularly, coordinated technical work with project partners, produced documentation and published at two international conferences; third paper in preparation.

TotalEnergies OneTech

Palaiseau, France

MACHINE LEARNING R&D ENGINEER — TRANSFER LEARNING — *Mar. 2023 — Sep. 2023*

- Led a transfer-learning study to improve multi-horizon wind-power forecasting across wind farms with heterogeneous data distributions.
- Conducted the literature review, time-series preparation, comparative experiments and interpretation; presented findings to R&D teams at Mathias Days.

ResilEyes Therapeutics

Paris, France

MACHINE LEARNING R&D INTERN — *Feb. 2022 — Jul. 2022*

- Developed machine-learning approaches on EEG signals to support the diagnosis, recognition and monitoring of post-traumatic stress disorder.
- Covered data collection and preprocessing, state-of-the-art review, prototyping, evaluation and results presentation.

Marc Jeannerod Institute of Cognitive Science

Bron, France

RESEARCH INTERN — *Jul. 2021 — Aug. 2021*

- Benchmarked and selected a framework for automated brain-tumour segmentation in 3D MRI under the institute's scientific and technical constraints.

Selected publications

2025	**Multi-cell Outdoor Channel State Information Dataset (MOCSID)**, M. E. M. Makhlouf, M. Guillaud, Y. Vindas — EuCNC & 6G Summit	*Poznań, Poland*
2025	**Charting Channels in the Presence of RIS**, M. E. M. Makhlouf, Y. Vindas, A. Kumar, M. Guillaud, M. Di Renzo — Asilomar Conference	*Pacific Grove, USA*

Research in progress

2026 **Controlling RIS-Induced Variability in Channel Charting for RIS Control**, First author - manuscript in preparation

Education

INSA Lyon · Inria Lyon *Lyon area, France*
PHD IN COMPUTER SCIENCE - SELF-SUPERVISED LEARNING FOR WIRELESS COMMUNICATIONS *2024 - Feb. 2027*
- Self-supervised representation learning for radio networks: channel charting, RIS, latent-dynamics prediction and privacy.

Université Paris Cité (Paris Cité University) *Paris, France*
MSC IN MATHEMATICS, MODELLING AND MACHINE LEARNING *2022 — 2023*
- Optimisation, statistical learning, image processing, inverse problems, probabilistic modelling and biostatistics.

Université Claude Bernard Lyon 1 *Lyon, France*
MSC IN COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE — WITH HONOURS *2020 — 2022*
- Machine learning, deep learning, reinforcement learning, multi-agent systems, image processing and parallel computing.

Université Claude Bernard Lyon 1 *Lyon, France*
BSC IN COMPUTER SCIENCE — WITH HONOURS; SECOND-BEST UNDERGRADUATE DISSERTATION *2018 — 2020*

Skills

AI and modelling	Self-supervised learning, deep learning, transfer learning, robust learning, machine unlearning, representation learning
Signals and data	Signal processing, time series, wireless CSI, computer vision, EEG, data preparation, statistics and visualisation
Programming	Python, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas; C/C++, R, Bash and SQL
Engineering	Git, Linux, HPC (Slurm/OAR, Grid'5000, Jean Zay), HDF5, API design and technical documentation
Research and delivery	Experimental design, reproducibility, technical writing, presentation, project coordination and EU collaboration
Languages	Arabic: native · French: fluent · English: professional · Berber: fluent · German: A1.1

Awards

2026 **3rd place — ESIT-D2I: Charting the Path to Bifröst**, Robust CSI localisation and machine unlearning under realistic data corruption *IEEE ITSoc / Kaggle*
2026 **1st place — scientific challenge**, INSTINCT Summer School, Castelldefels *Spain*

Community involvement

Volunteering Regular blood donor · Outreach and solidarity distributions with Secours populaire

Interests

Activities Independent travel · cycling and bicycle touring · hiking and climbing · multidisciplinary scientific curiosity